Convoy Management System

Database data flow | Database: `conveydb` | 4 November 2026

This document shows which database tables each action in the application reads and writes, from citizen registration through to trip arrival. It is based on the application models and on the create and update operations in the backend service.

1. How the main tables connect

```
ACCOUNTS
users ──────► registration_tbl (reg_id)           citizen and SCS
policeuser ─► police_registration_tbl (reg_id)    police and SP
admin                                             stands alone

TRIP (owner is registration_tbl.reg_id)
registration_tbl ─► vehicle_status_tbl ─► vehicle
                 ─► driver_status_tbl  ─► driver_tbl
                 ─► trip_tbl ─► trip_relation_tbl ─► passenger_tbl
                       │
                       ├─► approve_trip    (approved or rejected, by whom)
                       ├─► verified_trip   (verification record)
                       └─► checkout_trip   (arrival result)

CONVOY (per checkpost, per day)
origin_destination ─► tconvey ─► convey_control    departure start/stop
                              └► checkout_control  arrival start/stop
```

`reg_id` is the key to follow: it links an account to its vehicles, drivers and trips, and it records which officer approved a trip.

2. Normal trip, step by step

Step	Table	What happens
Register	`users`	New row with `isActive=0`
	`registration_tbl`	Personal details, new `reg_id`
	`auth_otp`	Hashed OTP
Verify OTP	`auth_otp`	`used=1`
	`users`	`isActive=1`
	`registration_tbl`	Updated
Add vehicle	`vehicle`, `vehicle_status_tbl`	Vehicle details, and the link to `reg_id`
Add driver	`driver_tbl`, `driver_status_tbl`	Driver details, and the link to `reg_id`
Create trip	`trip_tbl`	`status=1`, `verifiystatus=0`
	`passenger_tbl`, `trip_relation_tbl`	Passengers, and their link to the trip
	The same tables again	Second trip, if a return trip was chosen

Step	Table	What happens
Start convoy	`convey_control`	`status=1`, start time
Approve trip	`approve_trip`	`astatus=1`
	`trip_tbl`	`status=2`, `verifiystatus=2`
	`verified_trip`	`status=1`
Reject trip	`approve_trip`	`astatus=0`
	`trip_tbl`	`status=3`
	`verified_trip`	`status=0`
Stop convoy	`convey_control`	`status=0`, close time
Start/stop arrival	`checkout_control`	Opens and closes the arrival session
Record arrival	`checkout_trip`	1 Checked OK, 0 Non-arrival, 2 Problem
	`approve_trip`	`astatus=2`

> Recording an arrival does not change `trip_tbl.status`, and `astatus` becomes 2 even for Non-arrival. Use `checkout_trip.status` for the real result.

3. Special convoy (one save writes everything)

Special convoy trips need no police approval, so a single save writes all of these rows:

Table	What is written
`convey_control`	Convoy number 100+ (Emergency) or 200+ (VIP), the chosen time, `status=0`
`trip_tbl`	`status=2`, `verifiystatus=2`, convoy number in `convoyTime`
`passenger_tbl`, `trip_relation_tbl`	Passengers, if any
`approve_trip`	`astatus=1`, "Auto-approved"
`verified_trip`	`status=1`, "Auto-verified"

The arrival step is the same as for a normal trip: `checkout_trip`, then `approve_trip.astatus=2`.

4. Status values

Column	Values
`trip_tbl.status`	1 Pending, 2 Approved, 3 Rejected
`approve_trip.astatus`	0 Rejected, 1 Approved, 2 Arrival recorded
`checkout_trip.status`	0 Non-arrival, 1 Checked OK, 2 Problem
`convey_control.status`	1 Running, 0 Closed
Convoy number	1–99 Normal, 100–199 Emergency, 200 and above VIP

5. Accounts and security

Action	Table and effect
Captcha and CSRF	`sessions`
Wrong password	`failedLoginAttempts` and `lockedUntil` on the account row
Login or logout	`tokenVersion` on the account row, which cancels old tokens
Forgot password	`auth_otp`, then the new password on the account row
Freeze citizen	`users.isFrozen=1`
Freeze police	`policeuser.status=0`, plus `police_registration_tbl`
Register police (admin)	`policeuser`, `police_registration_tbl`
Security actions	`audit_log`

6. Deletes keep the data

Record	What a delete does
Vehicle	`vehicle.status=0`
Driver	`driver_tbl.dStatus='inactive'`
Trip	The `trip_tbl` row is updated, not removed
Police	`policeuser` and `police_registration_tbl` are updated, not removed

7. Other tables

- **Lookup tables (read only by the flows above):** `origin_destination` (checkposts), `tconvey`, `districts`, `subdistricts`, `villages`, `nationality_master`, `tdesignation`.
- **Trigger:** `before_trip_verifystatus_update` runs when `trip_tbl` is updated. A code comment says it sets `verifiedtime`; the trigger's SQL was not reviewed.
- **No longer written:** `otp`, replaced by `auth_otp`.
- **Probably unused:** `approval_relation`, `dummy_trip` and `states` have no model in the code. Check for raw SQL before removing them.
- **Trip edits:** `editTripDetailsByTripId` also changes `trip_tbl`, `passenger_tbl` and `trip_relation_tbl`; its details are not traced here.